

# 1141 CS140 資訊概論 Assignment 3

作業可適度使用 AI 大型語言模型協助理解與解題，但務必親自操作並確實理解指令的執行過程。將作業(用中英文都可)撰寫於 .doc 或 .pdf 格式的文件中，並將執行畫面截圖附於文件內一併呈現。

## 1. **grade.bat** — Student ID and Pass/Fail Checker (20 points)

Create a batch script named `grade.bat` that performs the following tasks:

- Step 1: Display your school ID and name using the `echo` command.
- Step 2: Prompt the user to enter a number between 0 and 100.
- Step 3: Check the input value:
  - If the number is greater than or equal to 60, display Pass.
  - If the number is less than 60, display Fail.

## 2. **ntime.bat** — Random Number Generator with Accumulated Sum (20 points)

Create a batch script named `ntime.bat` that performs the following tasks:

- Step 1: Every 10 seconds, generate a random number between 1 and 9.
- Step 2: After each number is generated:
  - Display the number.
  - Add it to an accumulated total.
  - Display the updated total.
- Step 3: Continue this process until the accumulated sum exceeds 100, then terminate the program.

3. **Command Behavior Comparison in Windows**— Please use both Command Prompt (CMD) or PowerShell in a Windows environment to execute the following commands. Observe their behavior and record the results. (50 points)

### ◆ CMD Commands:

- `timeout /t 10`
- `powershell -Command "Start-Sleep -Seconds 10"`

### ◆ PowerShell Commands:

- `timeout /t 10`
- `Start-Sleep -Seconds 10`

Compare the differences between these commands based on the following aspects:

- a) Does the command display any prompt or message during execution?
- b) Can the command be interrupted by the user (e.g., by pressing a key)?
- c) What additional insights or observations did you gain from this practice?